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Reinforcement Learning-Enabled Terrain Adaptation for Bipedal Wheeled-Legged Robots with Height Adjustment Capability

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***Abstract*—For the bipedal wheeled-legged robot (B-WLR), maintaining posture adjustment while adapting to complex terrains remains a key challenge to its broader deployment. To overcome this challenge, this study presents a reinforcement learning-based control framework that integrates terrain adaptability with posture regulation ability. By defining a mapping between the commanded body height and joint angles, also incorporating an active linear interpolation method, the robot gains the capability to adjust its height across varying terrains while maintaining stability. Meanwhile, to ensure efficient terrain traversal, an active evaluation mechanism is designed based on discrepancies among body velocity, commanded forward velocity, and wheel speed. Through this mechanism, the B-WLR lifts its legs when encountering vertical obstacles while relying on wheeled locomotion on other terrains, thereby enhancing adaptability to complex environments. The proposed strategy is validated through Webots simulations and experimentally verified on the LimX Dynamics Tron1.**

***Index Terms*—Wheel-legged robot, Reinforcement learning, Terrain adaptation, Motion control.**

I. Introduction

The wheel-legged drive (WLD) architecture, which integrates the advantages of both wheeled and legged locomotion, enables efficient mobility in complex environments through adaptive gait transitions. Bipedal wheeled-legged robots (B-WLRs) exhibit superior flexibility and maneuverability compared with other forms of WLDs. However, this configuration faces significant challenges when deployed on terrains with pronounced elevation variations. Although B-WLRs perform reliably on flat surfaces, they often lack sufficient adaptability to surmount unstructured terrains [1, 2]. In addition, most existing control frameworks depend on static gaits or predefined motion sequences, which further restrict their ability to cope with diverse terrain conditions [3].

On uneven surfaces, common approaches generally fail to correct the robot's posture or ground clearance in real-time,

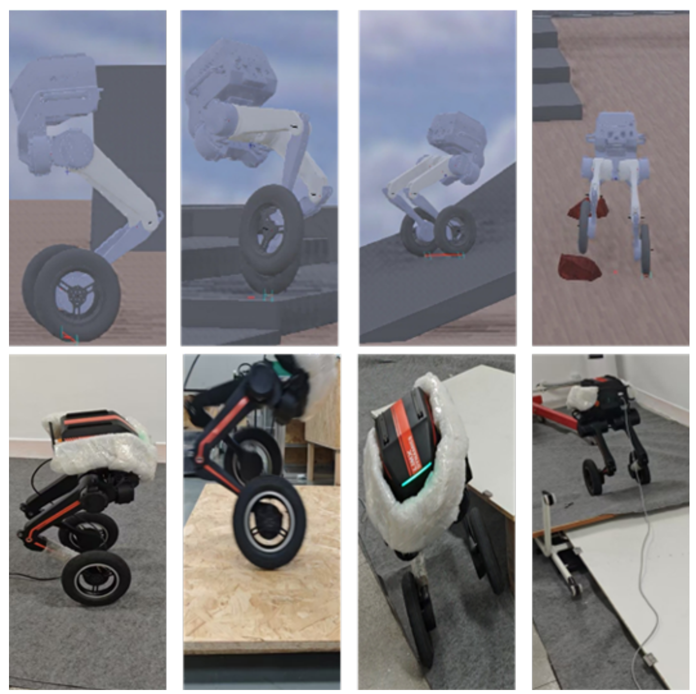

Fig. 1. The first row, from left to right, presents the Webots simulations: adjusting height, climbing stairs, ascending slopes, and traversing rough terrain. The second row, from left to right, shows the corresponding experimental validations. Supplementary video can be found at: https://drive.google.com/file/d/1FsoPUZlac7x1-Ce8O_E8-E4OALCyg5IT/view?usp=sharing

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基于强化学习的双轮腿机器人地形适应与高度调节能力

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摘要——对于双轮腿机器人（B-WLR）而言，在适应复杂地形的同时保持姿态调整，仍是其更广泛应用的关键挑战。为克服这一挑战，本研究提出了一种基于强化学习的控制框架，将地形适应性与姿态调节能力相结合。通过定义指令机体高度与关节角度之间的映射，并结合主动线性插值方法，机器人能够在不同地形上调节高度并保持稳定。同时，为确保高效的地形穿越，基于机体速度、指令前进速度和轮速之间的差异，设计了一种主动评估机制。通过该机制，B-WLR在遇到垂直障碍物时抬腿，而在其他地形上则依靠轮式运动，从而增强了对复杂环境的适应性。所提出的策略通过Webots仿真进行了验证，并在LimX Dynamics Tron1上进行了实验验证。

索引词——**轮腿机器人，强化学习，地形适应，运动控制。**

I. 引言

轮腿驱动（WLD）架构融合了轮式与腿式运动的优势，能够通过自适应步态转换在复杂环境中实现高效机动。双轮腿机器人（B-WLRs）相比其他形式的轮腿驱动系统展现出更优越的灵活性和可操控性。然而，当地形存在显著高度变化时，这种构型面临重大挑战。尽管双轮腿机器人在平坦表面上表现可靠，但它们往往缺乏足够的适应性为了克服非结构化地形 [1, 2]。此外，大多数现有控制框架依赖于静态步态或预定义运动序列，这进一步限制了它们应对多样化地形条件的能力 [3]。

在不平整表面上，常见方法通常无法实时纠正机器人姿态或离地间隙，

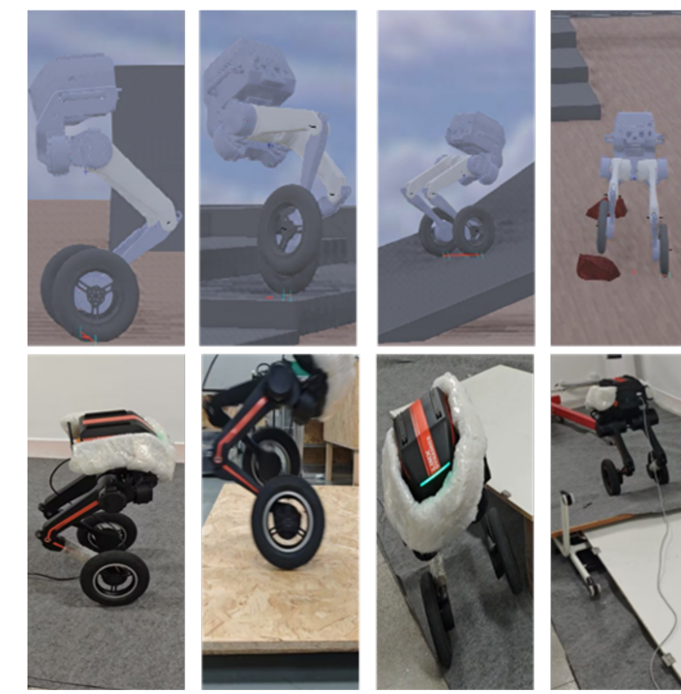

图1. 第一行从左至右展示了Webots仿真：高度调节、爬楼梯、上坡以及穿越崎岖地形。第二行从左至右展示了相应的实验验证。补充视频可访问：https://drive.google.com/file/d/1FsoPUZlac7x1-Ce8O E8-E4OALCyg5IT/view?usp=sharing

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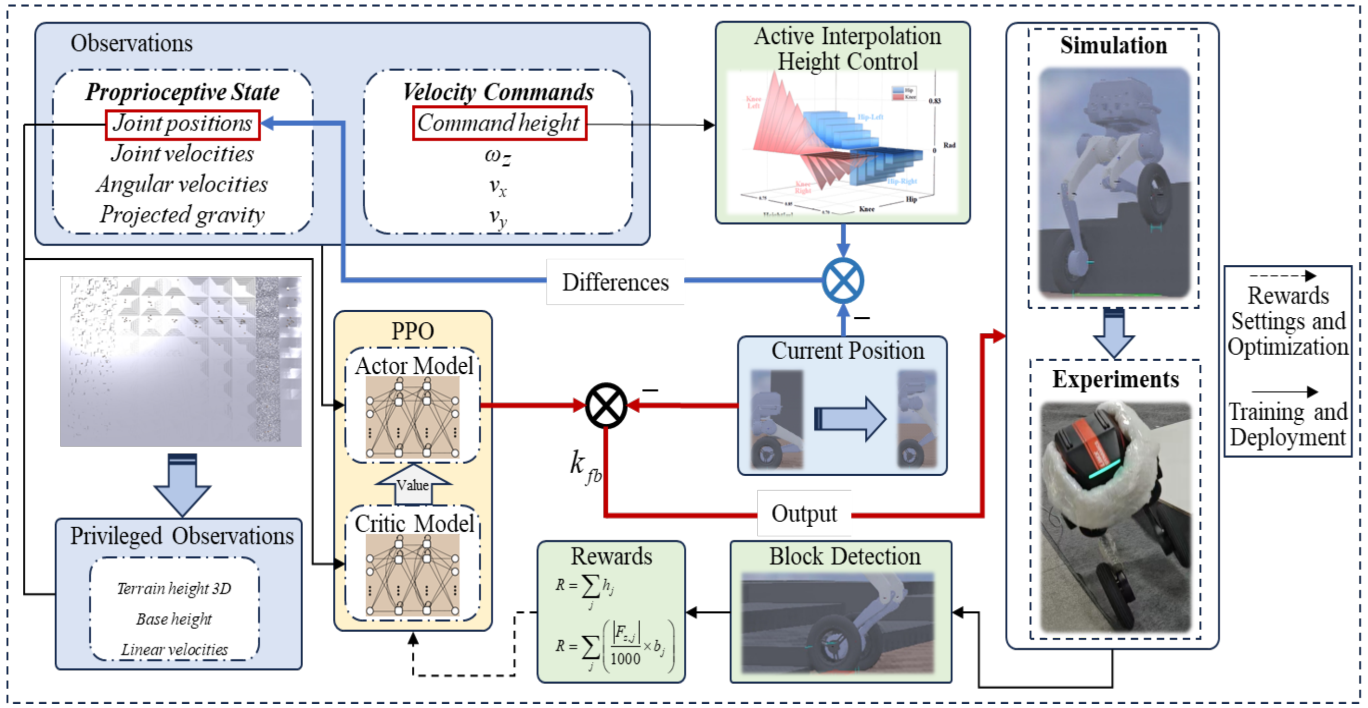

Fig. 2. Framework for the height control of a robotic system using PPO. The system receives proprioceptive states and privileged observations to generate velocity commands. The PPO model, consisting of an actor and a critic network, processes these inputs to provide active interpolation for height control. The system is validated in both simulation and real-world scenarios, with the performance evaluated in terms of rewards. Output:After optimization using the PPO algorithm, the difference between the position output by the Actor model and the current position is taken as the control output. Differences:The difference is defined as the deviation between the interpolated thigh and knee joint angles and those of the robot's current configuration.

thereby reducing its ability to traverse obstacles smoothly. Consequently, improving height adjustment capability while maintaining traversal capability has become a central issue in current research on wheeled-legged robots [4]. Recently, growing research efforts have focused on tackling the motion control challenges of B-WLRs in complex environments, with an emphasis on achieving active and precise body posture regulation while preserving efficient terrain traversal capability.

Conventionally, control methods for similar structures can be categorized into two parts [5]. One is the model-based control method, while the other is the data-driven approach, which has increasingly adopted reinforcement learning (RL) techniques in recent years [6].

In model-based control, Wensing *et al.* [7] provided a comprehensive review of model predictive control (MPC) and its applications in legged robots, noting that its performance is highly dependent on model accuracy. Bjelonic *et al.* [8] proposed a whole-body MPC framework combined with an online gait sequence generation method for wheeled-legged robots, enabling automatic transitions between wheel and gait patterns. To further enhance adaptability to unstructured terrains. Mason *et al.* [9] developed an MPC algorithm based on optimized contact forces, improving the robot's balance and driving capability on uneven surfaces. In conventional algorithms, precise posture regulation and multi-terrain traversal rely heavily on awareness of both the robot's internal states and external disturbances [10]. Agarwal *et al.* [11] introduced a state estimation method based on the extended Kalman filter, laying the foundation for reliable motion control. To address disturbances caused by load variations, Liu *et al.* [12] designed a load perception and localization framework for heavy-duty quadruped robots without external force sensors. Minniti *et al.* [13] integrated an adaptive control Lyapunov function with MPC, allowing the robot to estimate and compensate for unknown load dynamics online while maintaining stable motion.

RL has demonstrated exceptional robustness in controlling complex dynamical systems, providing an effective paradigm for achieving both terrain adaptability and precise posture regulation in B-WLRs. Hwangbo *et al.* [14] demonstrated the capability of RL to acquire dynamic locomotion skills in legged robots with high efficiency. Rudin *et al.* [15] further advanced this by introducing large scale GPU-based parallel simulations, which improved training efficiency even more and allowed robots to acquire walking skills within minutes, achieving seamless zero-shot transfer to real-world platforms. Building on this foundation, they introduced an end-to-end task framework guided by positional objectives, endowing robots with autonomous path planning and adaptive gait generation for efficient obstacle traversal [16].

Lee *et al.* [17] employed RL to train a quadruped wheeled-legged robot, enabling online transitions between driving mode and walking mode in response to the user command and terrain. Chamorro *et al.* [18] proposed an RL framework for

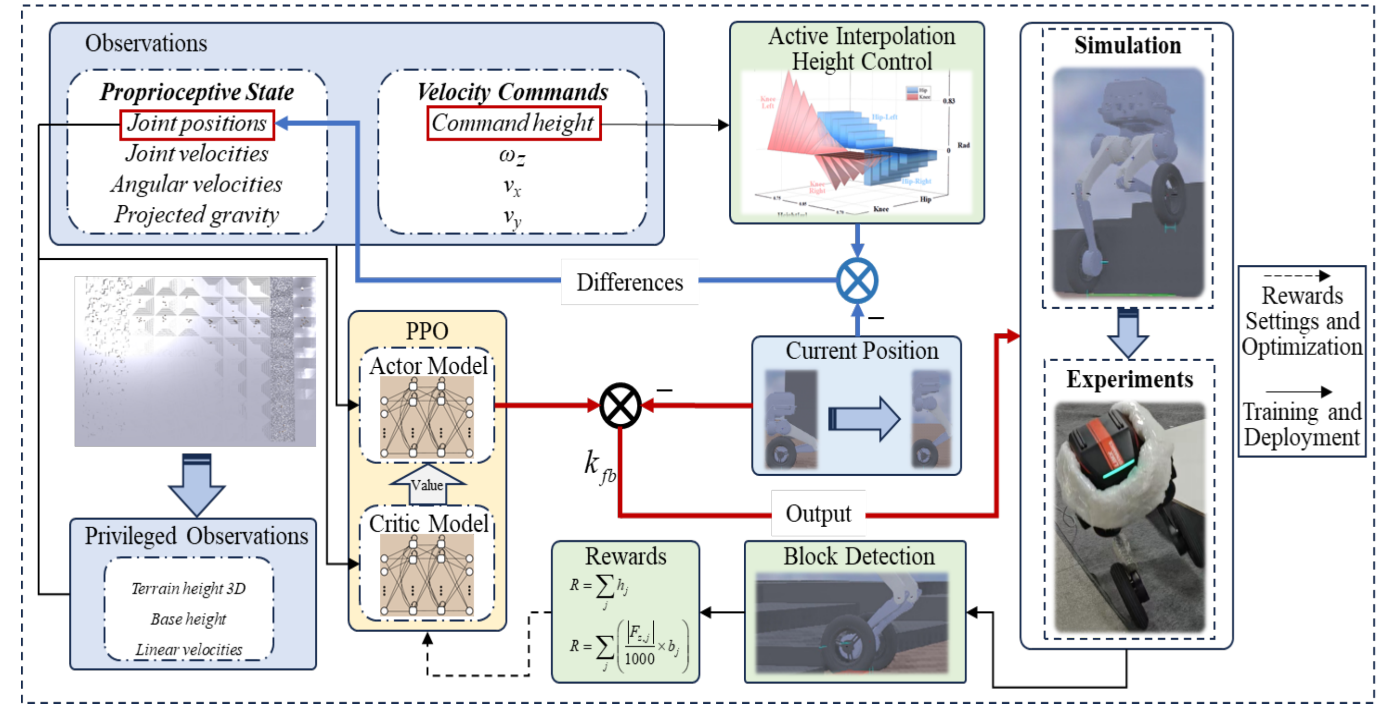

图2. 使用PPO的机器人系统高度控制框架。系统接收本体感受状态和特权观测以生成速度指令。由演员和评论家网络组成的PPO模型处理这些输入，为高度控制提供主动插值。该系统在仿真和真实场景中均得到验证，并通过奖励评估性能。输出：使用PPO算法优化后，演员模型输出的位置与当前位置之间的差值作为控制输出。差值：定义为插值后的大腿和膝关节角度与机器人当前构型角度之间的偏差。

从而降低了其平稳穿越障碍物的能力。因此，在保持越障能力的同时提升高度调节能力，已成为当前轮腿机器人研究的核心问题 [4]。近年来，越来越多的研究致力于解决B-WLR在复杂环境中的运动控制挑战，重点是在保持高效地形穿越能力的同时，实现主动且精确的身体姿态调节。

传统上，类似结构的控制方法可分为两类 [5]。一类是基于模型的控制方法，另一类是数据驱动方法，近年来数据驱动方法越来越多地采用强化学习（RL）技术 [6]。

在基于模型的控制中，Wensing等人 [7] 对模型预测控制（MPC）及其在腿式机器人中的应用进行了全面综述，指出其性能高度依赖于模型精度。Bjelonic 等人 [8]提出了一种全身MPC框架，结合在线步态序列生成方法，用于轮腿机器人，实现了轮式与步态模式之间的自动切换。为了进一步增强对非结构化地形的适应性，Mason等人 [9]开发了一种基于优化接触力的MPC算法，提升了机器人在不平整表面上的平衡与驱动能力。在传统算法中，精确姿态调节和多地形穿越高度依赖于对机器人内部状态和外部干扰的感知 [10]。Agarwal等人 [11]引入了一种基于扩展卡尔曼滤波器的状态估计方法，为可靠的运动控制奠定了基础。为了解决负载变化引起的干扰，Liu 等人 [12] 为重型四足机器人设计了一种无需外部力传感器的负载感知与定位框架。Minniti 等人 [13]将自适应控制李雅普诺夫函数与MPC相结合，使机器人能够在保持稳定运动的同时，在线估计并补偿未知的负载动力学。

强化学习在控制复杂动力系统方面展现出卓越的鲁棒性，为B-WLRs实现地形适应性和精确姿态调节提供了有效范式。Hwangbo 等人 [14]证明了强化学习能够高效获取腿式机器人的动态运动技能。Rudin 等人 [15]进一步推进了这项工作，引入大规模基于GPU的并行仿真，进一步提升了训练效率，使机器人能在数分钟内习得行走技能，并实现向真实平台的无缝零样本迁移。在此基础之上，他们引入了一种由位置目标引导的端到端任务框架，赋予机器人自主路径规划和自适应步态生成能力，以实现高效的障碍穿越 [16]。

Lee 等人 [17]采用强化学习训练了一款四轮腿式机器人，使其能够根据用户指令和地形在驾驶模式与行走模式之间进行在线切换。Chamorro 等人 [18] 提出了一种强化学习框架，用于

B-WLRs that integrates positional objectives with Boolean terrain-switching logic, allowing the robot to autonomously engage a stair-climbing mode and ascend 15*cm* steps using only onboard perception. Li *et al.* [19] introduced a contact-triggered blind climbing framework, where wheel-obstacle contact forces initiate feedforward leg-lifting trajectories, allowing Tron1 to surmount obstacles exceeding their wheel-hub radius. Lee *et al.* [20] developed an integrated control framework combining RL, privileged learning, and hierarchical control, achieving seamless transitions between walking and driving modes in complex tasks.

Although the aforementioned studies have achieved promising results, existing control frameworks frequently focus primarily on static adjustment or emphasize terrain traversal performance alone, making it difficult to balance terrain adaptability and posture regulation [3], which constrained the B-WLR's adaptability in practical applications. This study proposes an RL-based control framework that integrates terrain traversal capability with active posture regulation. The main contributions are as follows:

- 1) A velocity-triggered RL framework is proposed, in which the obstacle detection is achieved through the computation of a tri-source velocity error(referred to Sec.II-D). By incorporating an active leg-lifting strategy, the B-WLR attains efficient terrain traversal capability across diverse environments.
- 2) A command height-joint angle mapping is established to maintain terrain traversal performance. By applying piece-wise linear interpolation, the proposed framework enables rapid and adaptive posture adjustment.
- 3) The proposed RL framework is successfully deployed on the Tron1 robot. It demonstrates the ability to traverse 10*cm* stairs, a smooth slope, and dynamically perturbed inclines, while maintaining effective height regulation throughout these challenging terrains.

Fig.1 shows the simulation and experiments of the proposed framework. The rest of this paper is organized as follows: Sec.II shows the methodology, Sec III demonstrates the verifications, and Sec IV presents the conclusions.

II. Methodology

Fig.2 illustrates the RL framework used for training and deployment. In this subsection, we provide a description of the training learning environment, the RL task, the active interpolation height control, and the velocity-triggered strategy.

A. Learning Environment

1) Simulator and Platform: This study utilizes Isaac Gym for training and Webots for validation. During training, Isaac Gym's powerful parallel computing capabilities [21] enable the simultaneous execution of multiple environment instances on a single GPU, boosting policy learning efficiency. In the validation phase, Webots is used for high-fidelity simulation [22], offering a fast and efficient platform for verifying trained policies. Cross-validation is performed using the open-source Tron1 Unified Robot Description Format (URDF) model in Webots [23], ensuring policy adaptability across various simulation environments. This process guarantees the reliable transfer of policies from simulation to real-world applications.

2) Learning Algorithm: The research employs a learning framework that combines Proximal Policy Optimization (PPO) with an asymmetric actor-critic architecture. Within this framework, the critic network performs value estimation by leveraging privileged information, then transmits the derived value to the actor network to guide its policy optimization. All training was conducted on a single NVIDIA RTX 4090 GPU, with convergence achieved within 6,000 iterations, demonstrating the algorithm's high efficiency.

B. Task Formulation

1) State: In this study, we adopt an asymmetric actor-critic framework, where the state is divided into observation and privileged information. The observation is available to both the actor and critic during training and deployment, while the privileged information is accessible only to the critic during training. Their detailed definitions are listed in Tables.I and II. The joint positions are the knee angle after active interpolation, which can be inferred from Fig.3.

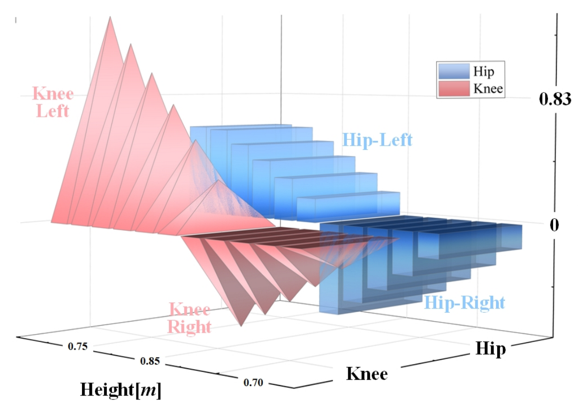

Fig. 3. Active interpolation data during the training, and the corresponding posture can be found in Fig.4

The aforementioned method effectively reflects the robot's relative position changes in the environment, particularly in complex or dynamic terrains. Through active linear interpolation control, the B-WLR can dynamically adjust according to predefined control objectives, ensuring that the height-joint position adapts to the real-world environment during task execution. Due to the difficulty of directly obtaining the robot's linear velocity and absolute base height of a B-WLR, this data is provided as privileged information and excluded from the actor network's observations. Additionally, the training environment supplies actual terrain height data to the privileged information, enabling the adjustment of relative height across various terrains.

2) Actions: In this study, the active numbers of joints of the tested robot are 8 (abads, hips, knees, and wheels), and the output dimension of the action was set to 8.

The actions command can be used as input to the PD controller, allowing the preset controller to adjust the joint status of the robot to achieve the target posture. As for the

B-WLRs，该框架将位置目标与布尔地形切换逻辑相结合，使机器人能够自主进入爬楼梯模式，并仅依靠机载感知攀爬15厘米高的台阶。Li 等人 [19] 提出了一种接触触发的盲攀爬框架，其中轮子与障碍物的接触力会触发前馈抬腿轨迹，使Tron1能够越过超过其轮毂半径的障碍物。
Lee 等人 [20] 开发了一种集成控制框架，该框架结合了强化学习、特权学习和分层控制，在复杂任务中实现了行走模式与驾驶模式之间的无缝切换。

尽管上述研究已取得令人鼓舞的成果，但现有控制框架往往主要侧重于静态调节或仅强调地形穿越性能，难以平衡地形适应性与姿态调节 [3]，，这限制了B-WLR在实际应用中的适应性。本研究提出了一种基于强化学习的控制框架，该框架将地形穿越能力与主动姿态调节相结合。主要贡献如下：

- 1) 提出了一种速度触发的强化学习框架，其中通过计算三源速度误差（详见第II-D节）实现障碍物检测。通过引入主动抬腿策略，B-WLR在不同环境中获得了高效的地形穿越能力。
- 2) 建立了指令高度与关节角度的映射关系，以维持地形穿越性能。通过应用分段线性插值，所提出的框架能够实现快速且自适应的姿态调节。
- 3) 所提出的强化学习框架已成功部署于Tron1机器人上。它展示了穿越10厘米楼梯、平滑斜坡以及动态扰动坡面的能力，同时在这些具有挑战性的地形上保持了有效的高度调节。

图1展示了所提出框架的仿真与实验。本文其余部分组织如下：第二节介绍方法论，第三节展示验证结果，第四节给出结论。

II. 方法论

图2 展示了用于训练和部署的强化学习框架。在本小节中，我们描述了训练学习环境、强化学习任务、主动插值高度控制以及速度触发策略。

A. 学习环境

1) 仿真器与平台：本研究采用Isaac Gym进行训练，并使用Webots进行验证。在训练阶段，Isaac Gym强大的并行计算能力 [21] 能够在单个GPU上同时执行多个环境实例，从而提升策略学习效率。在验证阶段，Webots用于高保真仿真[22], 为验证训练好的策略提供了一个快速高效的平台。交叉验证使用开源的Tron1统一机器人描述格式（URDF）模型在Webots [23], 中进行，确保策略在各种仿真环境中的适应性。这一过程保证了策略从仿真到现实世界应用的可靠迁移。

2) 学习算法：本研究采用了一种结合近端策略优化（PPO）与非对称演员-评论家架构的学习框架。在该框架中，评论家网络通过利用特权信息进行价值估计，然后将推导出的价值传递给演员网络以指导其策略优化。所有训练均在单个NVIDIA RTX 4090 GPU上进行，并在6000次迭代内实现收敛，证明了该算法的高效性。

B. 任务制定

1) 状态：在本研究中，我们采用非对称演员-评论家框架，其中状态分为观测和特权信息。观测在训练和部署期间对演员和评论家均可用，而特权信息仅在训练期间对评论家可访问。它们的详细定义列于表I 和II中。关节位置是经过主动插值后的膝关节角度，可从图3中推断得出。

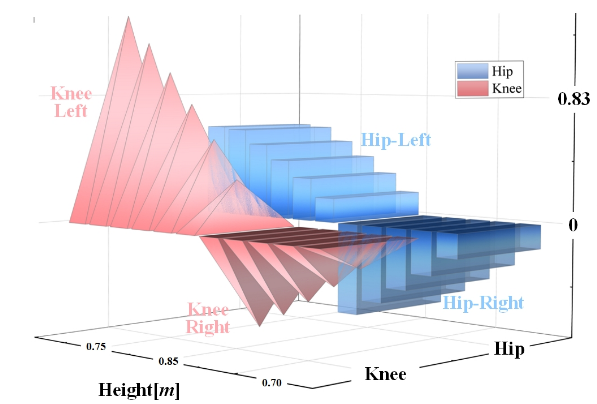

图3. 训练期间的主动插值数据，相应的姿态可在图4中找到。

上述方法有效反映了机器人在环境中的相对位置变化，尤其是在复杂或动态地形中。通过主动线性插值控制，B-WLR可以根据预定义的控制目标动态调整，确保高度-关节位置在任务执行过程中适应真实环境。由于直接获取B-WLR的线速度和绝对基座高度存在困难，这些数据作为特权信息提供，并从演员网络的观测中排除。此外，训练环境向特权信息提供实际地形高度数据，从而能够在各种地形上调整相对高度。

2) 动作：在本研究中，被测机器人的主动关节数量为8个（abad、髋关节、膝关节和轮子），动作的输出维度设置为8。

动作指令可作为PD控制器的输入，使预设控制器能够调整机器人的关节状态，从而实现目标姿态。至于

TABLE I
OBSERVATION

Symbols	Name	Units	Coeff.	Size	Noise(%)
$\vec{q}$	Joint positions	*rad*	1.0	6	±1
$\dot{\vec{q}}$	Joint velocities	*rad/s*	0.05	8	±150
$\dot{\vec{\theta}}$	Angular velocity	*rad/s*	0.25	3	±20
$\vec{\gamma}$	Projected gravity	-	1.0	3	±5
h_{target}	Height command	*m*	0.8	1	±0

TABLE II
PRIVILEGED INFORMATION

Symbols	Name	Units	Coeff.	Size	Noise(%)
H_{terrain}	Terrain height	*m*	5.0	187	-
h	Base height	*m*	0.8	31	-
v_x	Linear velocity	*m/s*	2.0	3	-

TABLE III
REWARDS

#	Reward	Formula	Value
1	Tracking lin vel	$\exp\left(-\frac{\lVert \mathbf{c}_v - \mathbf{v}_{\text{base}} \rVert^2}{\sigma_t^2}\right)$	2
	Tracking ang vel	$\exp\left(-\frac{(c_\omega - \omega_{\text{base}})^2}{\sigma_t^2}\right)$	1.6
	Lin vel error	$\lVert \mathbf{c}_v - \mathbf{v}_{\text{base}} \rVert^2$	-1
	Lin vel z	v_z^2	-0.05
	Ang vel xy	$\omega_x^2 + \omega_y^2$	-0.01
2	Collision	$\sum_{j \in \mathcal{I}_{\text{peralty}}} \mathbb{I}(\lVert \mathbf{F}_{c,j} \rVert > 0.1)$	-3.0
	Block interval	$R = \sum_j h_j$	-0.8
	Leg obstruction	$R = \sum_j \left(\frac{\lvert F_{z,j} \rvert}{1000} \times b_j\right)$	-0.5
3	Nominal foot position	$\left[\frac{1}{N}\sum_{i=1}^{N} \exp\left(-\frac{e_{h,i}^2}{\sigma_f^2}\right)\right] \times \exp\left(-\frac{v_{\text{cmd}}^2}{\sigma_v^2}\right)$	0.8
	Tracking height	$\exp\left(-\frac{(h_{\text{base}} - c_h)^2}{\sigma_h^2}\right)$	0.35
	Base height error	$(h_{\text{base}} - c_h)^2$	-8.0
4	Dof acc	$\sum_i \left(\frac{\dot{q}_{i,\text{ last}} - \dot{q}_i}{\Delta t}\right)^2$	-8e-7
	Dof pos limits	$\sum_i \left[\max\left(0, q_i^{\min} - q_i\right) + \max\left(0, q_i - q_i^{\max}\right)\right]$	-5.0
	Torque limits	$\sum_i \max\left(0, \lvert\tau_i\rvert - \tau_i^{\text{limit}} \cdot r_{\text{soft}}\right)$	-1.2
	Torque	$\sum_i \tau_i^2$	-2e-4
	Action rate	$\sum_i \left(a_{i,\text{ last}} - a_i\right)^2$	-0.04
5	Keep balance	1	1.2
	Stability recovery	$\begin{cases} \max(0, -\text{sign}(\phi)\cdot\dot{\phi}) + \max(0, -\text{sign}(\theta)\cdot\dot{\theta}) & \text{if } \lvert\phi\rvert > 0.09 \text{or} \lvert\theta\rvert > 0.09 \\ 0 & \text{otherwise} \end{cases}$	0.02
	No presquat	$\max\left(0, (c_h - h_{\text{base}}) - m\right) \times \mathbb{I}\left(\lvert v_{\text{cmd}} \rvert > v_{\text{gate}}\right)$	-0.05

wheel joint, it achieves the action control through the derivative control only, which means the position gains were set to zero during the motion control of the wheel. Additionally, we set the position gains for the joints and wheel individually to achieve the proper motion control.

3) Rewards: The main setting of the rewards is demonstrated through Table.III. The reward setting in this study was grant greater flexibility to traverse complex terrains, enabling solutions that are difficult for traditional algorithms to achieve. Reward #1 ensures the robot maintains a preset speed. Reward #2 promotes adaptive motion planning to avoid collisions in complex terrains. Reward #3 uses height and posture control to stabilize the B-WLR on uneven terrain. Reward #4 maintains reasonable joint torque and motion ranges during movement and height adjustment. Reward #5 improves training stability by preventing unwanted crouching behaviors without a forward command.

4) Curriculum: In the training environment, a total of 8192 agents are deployed in parallel to accelerate policy convergence. A curriculum learning strategy is adopted to gradually increase the task difficulty over ten progressive levels. The update policy is defined as follows:

When an agent successfully traverses more than 50% of the total terrain length, the terrain curriculum level is advanced, and additional command types are introduced.

Specifically, the command curriculum dynamically adjusts the range of control commands based on the tracking performance of key motion variables. When the reward for linear velocity tracking exceeds 90% of its maximum over an episode, the allowable range of linear velocity commands along the x-axis is expanded to encourage faster locomotion behaviors within the predefined ranges. Similarly, when the angular velocity tracking reward surpasses 70% of its maximum, the range of yaw angular velocity commands is increased to enhance turning agility and robustness.

In addition, if the commanded height approaches the limits of the current curriculum stage, the minimum and maximum height boundaries are progressively extended, thereby increasing the complexity of height regulation and improving adaptability to diverse terrains.

C. Active Interpolation Height Control

The preset robot height and joint inflection points were computed using MATLAB/Simscape. The corresponding visualized data were derived from the URDF file, as shown in Figs.3 and 4. By applying active linear interpolation, the joint positions associated with each target command were accurately determined. This process provides the simulated robot with a precise motion baseline, improving training efficiency. As a result, the robot can adjust its height accurately on complex terrain and maintain stability by avoiding improper leg-raising motions. Interpolation data as shown in Table.IV.

D. Velocity-triggered leg-lifting strategy

When operating on uneven terrain or encountering obstacles, the robot needs to adapt its motion to maintain effective locomotion. To address this, a velocity-triggered leg-lifting strategy is developed. It considers variations in wheel linear velocity, body velocity, and sustained forward command velocity obtained via the remote controller to trigger appropriate leg-lifting actions, which can help improve motion adaptability and overall locomotion efficiency. The leg-lifting triggered strategy is shown as

1) : When the robot receives a sustained forward command and its wheel linear velocity exceeds the body velocity, it is considered to have encountered an obstacle. In this case, an active leg-lifting action is triggered to facilitate obstacle negotiation.

2) : When both the wheels' linear velocities and body velocity approach zero while a forward command remains active, the system triggers a leg-lifting action. It helps resolve the robot stagnation issue, ensuring that the robot can continue effective locomotion even when encountering obstacles.

表一 观测

符号	Name	单位	系数	Size	噪声(%)
$\vec{q}$	关节位置	*rad*	1.0	6	±1
$\dot{\vec{q}}$	关节速度	弧度/秒	0.05	8	±150
$\dot{\vec{\theta}}$	角速度	弧度/秒	0.25	3	±20
$\vec{\gamma}$	投影重力	-	1.0	3	±5
h_{target}	高度指令	*m*	0.8	1	±0

表二 特权信息

符号	Name	单位	C系数	Size	噪声(%)
H_{terrain}	地形高度	*m*	5.0	187	-
h	基准高度	*m*	0.8	31	-
v_x	线速度	*m/s*	2.0	3	-

表III 奖励

#	奖励	公式	值
1	线速度跟踪	$\exp\left(-\frac{\lVert \mathbf{c}_v - \mathbf{v}_{\text{base}} \rVert^2}{\sigma_t^2}\right)$	2
	角速度跟踪	实验 $\left(-\frac{(c_\omega - \omega_{\text{基础}})^2}{\sigma_t^2}\right)$	1.6
	线速度误差	$\lVert \mathbf{c}_v - \mathbf{v}_{\text{base}} \rVert^2$	-1
	线速度Z分量	v_z^2	-0.05
	角速度 xy	$\omega_x^2 + \omega_y^2$	-0.01
2	碰撞	$\sum_{j \in \mathcal{I}_{\text{惩罚}}} \mathbb{I}(\lVert \mathbf{F}_{c,j} \rVert > 0.1)$	-3.0
	阻塞间隔	$R = \sum_j h_j$	-0.8
	腿部障碍	$R = \sum_j \left(\frac{\lvert F_{z,j} \rvert}{1000} \times b_j\right)$	-0.5
3	标称足部位置	$\left[\frac{1}{N}\sum_{i=1}^{N} \exp\left(-\frac{e_{h,i}^2}{\sigma_f^2}\right)\right] \times \exp\left(-\frac{v_{\text{cmd}}^2}{\sigma_v^2}\right)$	0.8
	跟踪高度	$\exp\left(-\frac{(h_{\text{base}} - c_h)^2}{\sigma_h^2}\right)$	0.35
	基座高度误差	$(h_{\text{base}} - c_h)^2$	-8.0
4	自由度加速度	$\sum_i \left(\frac{\dot{q}_{i,\text{最后}} - \dot{q}_i}{\Delta t}\right)^2$	-8e-7
	自由度位置限制	$\sum_i \left[\text{最大}\left(0, q_i^{\min} - q_i\right) + \text{最大}\left(0, q_i - q_i^{\max}\right)\right]$	-5.0
	扭矩限制	$\sum_i \text{最大}\left(0, \lvert\tau_i\rvert - \tau_i^{\text{limit}} \cdot r_{\text{soft}}\right)$	-1.2
	扭矩	$\sum_i \tau_i^2$	-2e-4
	动作速率	$\sum_i \left(a_{i,\text{ last}} - a_i\right)^2$	-0.04
5	保持平衡	1	1.2
	稳定性恢复	$\begin{cases} \max(0, -\text{sign}(\phi)\cdot\dot{\phi}) + \max(0, -\text{sign}(\theta)\cdot\dot{\theta}) & \text{如果 } \lvert\phi\rvert > 0.09 \text{或} \lvert\theta\rvert > 0.09 \\ 0 & \text{否则} \end{cases}$	0.02
	无预蹲	$\max\left(0, (c_h - h_{\text{base}}) - m\right) \times \mathbb{I}\left(\lvert v_{\text{cmd}} \rvert > v_{\text{gate}}\right)$	-0.05

轮关节，它仅通过微分控制实现动作控制，这意味着在轮子的运动控制过程中位置增益被设为零。此外，我们分别为关节和轮子设置了位置增益，以实现适当的运动控制。

3) 奖励： 奖励的主要设置通过表格III展示。本研究中的奖励设置为穿越复杂地形提供了更大的灵活性，从而实现了传统算法难以达成的解决方案。奖励#1确保机器人保持预设速度。奖励#2促进自适应运动规划，以避免在复杂地形中发生碰撞。奖励#3利用高度和姿态控制，使B-WLR在不平坦地形上保持稳定。奖励#4在移动和高度调节过程中维持合理的关节扭矩和运动范围。奖励#5通过防止在没有前进指令时出现不必要的蹲伏行为，提高了训练稳定性。

*4)*课程学习： 在训练环境中，共部署了8192个智能体并行运行，以加速策略收敛。采用课程学习策略，通过十个渐进级别逐步增加任务难度。更新策略定义如下：

当某个智能体成功穿越超过总地形长度50%时，地形课程级别将提升，并引入额外的指令类型。

具体而言，指令课程根据关键运动变量的跟踪性能动态调整控制指令的范围。当线速度跟踪奖励在一个回合内超过其最大值的90%时，x轴方向的线速度指令允许范围将被扩大，以鼓励在预定义范围内实现更快的运动行为。类似地，当角速度跟踪奖励超过其最大值的70%时，偏航角速度指令的范围将被增加，以提升转向敏捷性和鲁棒性。

此外，如果指令高度接近当前课程阶段的极限，则最小和最大高度边界会逐步扩展，从而增加高度调节的复杂性，并提升对多样化地形的适应性。

C. 主动插值高度控制

预设的机器人高度和关节拐点通过MATLAB/Simscape计算得出。相应的可视化数据源自URDF文件，如图3 和4所示。通过应用主动线性插值，与每个目标指令相关的关节位置被精确确定。这一过程为仿真机器人提供了精确的运动基准，提高了训练效率。因此，机器人能够在复杂地形上精确调整高度，并通过避免不当的抬腿动作来保持稳定性。插值数据如表IV所示。

D. 速度触发的抬腿策略

当机器人在不平坦地形上运行或遇到障碍物时，需要调整其运动以保持有效的移动能力。为此，开发了一种速度触发的抬腿策略。该策略考虑通过遥控器获取的轮子线速度、机体速度以及持续前进指令速度的变化，以触发适当的抬升动作，从而有助于提高运动适应性和整体移动效率。触发的抬腿策略如下所示

1) : 当机器人接收到持续的前进指令，且其轮子线速度超过机体速度时，视为遇到了障碍物。在这种情况下，会触发主动抬升动作，以帮助机器人越过障碍物。

2): 当轮子的线速度和机体速度均趋近于零，而前进指令仍处于激活状态时，系统会触发抬升动作。这有助于解决机器人停滞问题，确保机器人在遇到障碍物时仍能继续有效运动。

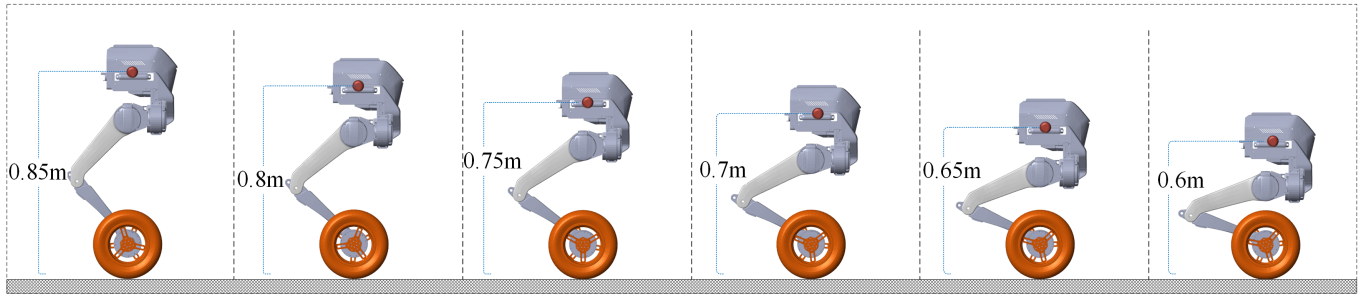

Fig. 4. Preset Height-Joint Angle Mapping Relationship.

TABLE IV
ACTIVE LINEAR INTERPOLATION DATA

Name	Data1	Data2	Data3	Data4	Data5	Data6
Height(m)	0.60	0.65	0.70	0.75	0.80	0.85
$\text{Hip}_\text{L}(rad)$	0.6083	0.5892	0.4938	0.3923	0.2823	0.1590
$\text{Knee}_\text{L}(rad)$	1.3606	1.1784	0.9875	0.7847	0.5646	0.3181

This mechanism automatically adjusts the leg-lifting motion by continuously monitoring the contact status, obstruction intensity, and motion parameters at the robot's wheel, thereby resolving potential stalling issues when the robot encounters obstacles.

Blocked detection The system first detects the contact state of each foot. If both feet are impeded, it evaluates the obstruction intensity and assigns a weight to each foot based on the contact and tangential forces, prioritizing the lifting action on the more heavily obstructed side.

Set $sat(u) = \tanh\left(\frac{u}{v_{\max}}\right)$ as the soft upper limit for forward velocity. Set weights:

$$w_{\text{both}}^{\uparrow},\ w_{\text{both}}^{\rightarrow},\ w_{\text{single}}^{\uparrow},\ w_{\text{single}}^{\rightarrow} > 0, \tag{1}$$

When both feet are obstructed and at nonsim heights, soft weight aggregation is employed:

$$R_{\text{nonsim}} = \sum_{i=1}^{2} w_i \left(w_{\text{both}}^{\uparrow} |v_{z,i}| + w_{\text{both}}^{\rightarrow} \operatorname{sat}\left(|v_{x,i}|\right) \right), \tag{2}$$

where, w_i is the soft weight calculated via the softmax function based on foot height. $w_{\text{both}}^{\uparrow}$ denotes the reward weight for vertical velocity when both feet are obstructed, while $w_{\text{both}}^{\rightarrow}$ represents the reward weight for forward velocity under the same condition.

When both feet are obstructed at similar heights or only one foot is obstructed, sum the obstructed feet independently:

$$R_{\text{sim}} = \sum_{i=1}^{2} b_i \left(w_{\text{single}}^{\uparrow} |v_{z,i}| + w_{\text{single}}^{\rightarrow} \operatorname{sat}\left(|v_{x,i}|\right) \right), \tag{3}$$

where, b_i is the indicator function for the obstruction state of the i-th leg, $w_{\text{single}}^{\uparrow}$ is the reward weight for vertical velocity in this scenario, and $w_{\text{single}}^{\rightarrow}$ is the reward weight for forward velocity.

When the forward stride length exceeds the limit, the robot will incur a stride penalty.

$$S_x = \max(x_1, x_2) - \min(x_1, x_2), \tag{4}$$

$$P_{\text{span}} = \lambda_{\text{span}} \left[S_x - S_x^{\max} \right]_+, \tag{5}$$

where$[u]_+ = \max(u, 0)$, S_x is the span between the two feet along the x-axis in the body coordinate system, $S_x^{\max}$ is the maximum allowable span threshold, and λ_{span} is the span penalty coefficient.

Therefore, the total reward is adjusted based on different scenarios, as shown in the following formula:

$$\begin{aligned} R =& 1\{\text{nonsim}\} R_{\text{nonsim}} \\ &+ 1\{\text{similar or } (b_1 + b_2 = 1)\} R_{\text{sim}} - P_{\text{span}}, \end{aligned} \tag{6}$$

Ascending Velocity Reward: Using the weights derived from the obstruction intensity, the system provides a reward when the vertical foot velocity is positive, encouraging the robot to continue lifting that foot.

Wheel Linear Velocity Penalty: If the wheel linear velocity exceeds a predefined threshold, the system applies a penalty to suppress excessive speed, ensuring smooth leg-lifting motions and preventing instability caused by abrupt movements.

The above triggering mechanisms ensured that the robot can flexibly adjust its motion strategy according to real-time block conditions, enabling more natural and stable locomotion on complex terrains.

III. EXPERIMENTS

As shown in Fig.5, the training consists of modular terrain units within an 8×8m area, featuring five distinct terrain configurations. Two columns are dedicated to staircase terrain, while the remaining three columns include smooth slope, rough slope, and terrain with discrete obstacles. Each terrain unit is divided into 10 progressively increasing difficulty levels: base difficulty + coefficient×difficulty. The difficulty is defined as [0.5, 0.75, 0.9].

By adding friction noise to the terrain height field, the irregularity of natural terrain is simulated. Different scales of noise are generated by adjusting parameters such as frequency, detail level, amplitude decay, and frequency growth, and then

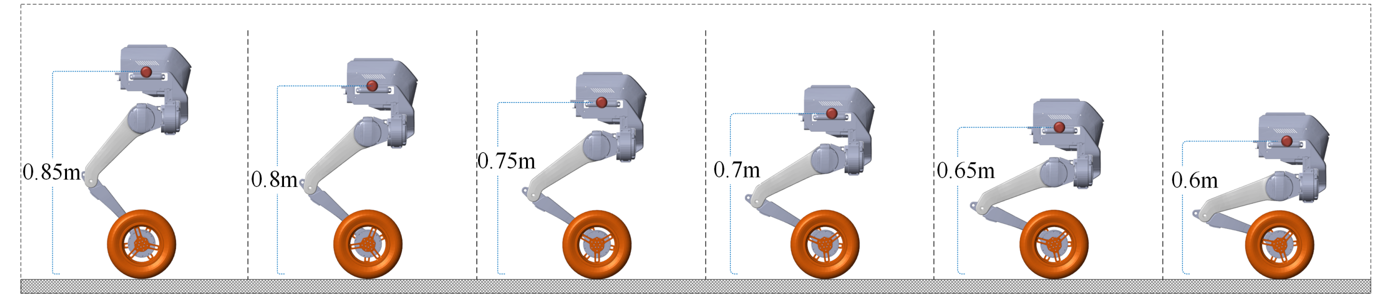

图4. 预设高度-关节角度映射关系。

表IV 主动线性插值数据

Name	数据1	数据2	数据3	数据4	数据5	数据6
高度(米)	0.60	0.65	0.70	0.75	0.80	0.85
髋关节$_\text{L}$(弧度)	0.6083	0.5892	0.4938	0.3923	0.2823	0.1590
膝关节$_\text{L}$(弧度)	1.3606	1.1784	0.9875	0.7847	0.5646	0.3181

该机制通过持续监测机器人轮子的接触状态、障碍强度及运动参数，自动调整抬腿运动，从而解决机器人遇到障碍物时可能出现的失速问题。

阻塞检测 系统首先检测每只脚的接触状态。如果双脚均受阻，则评估障碍强度，并根据接触态和切向力为每只脚分配权重，优先对障碍更严重的一侧执行抬升动作。

将 $sat(u) = \tanh\left(\frac{u}{v_{\max}}\right)$ 设为前进速度的软上限。设置权重：

$$w_{\text{both}}^{\uparrow},\ w_{\text{both}}^{\rightarrow},\ w_{\text{single}}^{\uparrow},\ w_{\text{single}}^{\rightarrow} > 0, \tag{1}$$

当双足均受阻且处于不同高度时，采用软权重聚合：

$$R_{\text{nonsim}} = \sum_{i=1}^{2} w_i \left(w_{\text{both}}^{\uparrow} |v_{z,i}| + w_{\text{both}}^{\rightarrow} \operatorname{sat}\left(|v_{x,i}|\right) \right), \tag{2}$$

其中， w_i 是基于足部高度通过 softmax函数 计算得到的软权重。 $w^{\uparrow}$ 双足均受阻时表示双足均受阻时垂直速度的奖励权重，而 $w_{\text{both}}^{\rightarrow}$ 表示相同条件下前进速度的奖励权重。

当双足在相似高度受阻或仅单足受阻时，独立累加受阻足部：

$$R_{\text{sim}} = \sum_{i=1}^{2} b_i \left(w_{\text{single}}^{\uparrow} |v_{z,i}| + w_{\text{single}}^{\rightarrow} \operatorname{sat}\left(|v_{x,i}|\right) \right), \tag{3}$$

其中， b_i 是第i条腿阻塞状态的指示函数， $w^{\uparrow}$ 单一是该场景下垂直速度的奖励权重， $w_{\text{single}}^{\rightarrow}$ 是前进速度的奖励权重。

当前进步长超过限制时，机器人将受到步幅惩罚。

$$S_x = \max(x_1, x_2) - \min(x_1, x_2), \tag{4}$$

$$P_{\text{span}} = \lambda_{\text{span}} \left[S_x - S_x^{\max} \right]_+, \tag{5}$$

其中 $[u]_+ = \max(u,0)$， S_x 是机体坐标系中双足沿 x 轴 的跨度， $S_x^{\max}$ 是最大允许跨度阈值， λ_{span} 是跨度惩罚系数。

因此，总奖励根据不同场景进行调整，如下式所示：

$$\begin{aligned} R =& 1\{\text{nonsim}\} R_{\text{nonsim}} \\ &+ 1\{\text{similar or } (b_1 + b_2 = 1)\} R_{\text{sim}} - P_{\text{span}}, \end{aligned} \tag{6}$$

上升速度奖励：利用从障碍强度导出的权重，当足部垂直速度为正时，系统提供奖励，鼓励机器人继续抬起该脚。

轮子线速度惩罚：如果轮子线速度超过预设阈值，系统会施加惩罚以抑制过快的速度，确保平滑的抬腿动作，并防止因突然运动导致的不稳定。

上述触发机制确保机器人能够根据实时阻塞条件灵活调整其运动策略，从而在复杂地形上实现更自然、更稳定的运动。

III. 实验

如图5所示，训练由 8×8米区域内的模块化地形单元组成，包含五种不同的地形配置。两列专用于楼梯地形，其余三列包括平滑斜坡、粗糙斜坡和离散障碍物地形。每个地形单元分为10个逐步递增的难度等级：基础难度 + 系数×难度。难度定义为 [0.5,0.75,0.9]。

通过在地形高度场中添加摩擦噪声，模拟了自然地形的不规则性。通过调整频率、细节层次、振幅衰减和频率增长等参数，生成不同尺度的噪声，然后

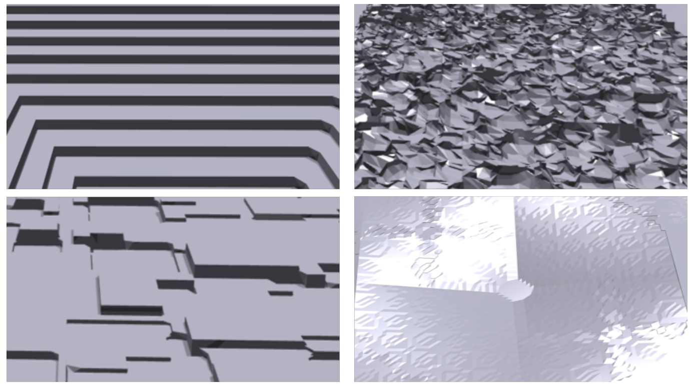

Fig. 5. First row left to right: stair terrain, discrete obstacle terrain. Second row: rough slope, smooth terrain.

overlaid onto the terrain. The generated noise enhances the terrain's variations, improving its realism.

Stair terrain: In staircase terrain design, step height increases with difficulty. Based on robot training performance, the step height is adjusted between $5cm$ and $10cm$. The step width is fixed at 0.35, with each independent platform dimension set to 2.0 meters. This simulates stair inclination, enabling the representation of stair terrain within a single function.

Rough slope terrain: In the design of rough slope terrains, the slope adjusts with increasing difficulty. The difficulty is governed by the size of the independent platform. When the platform size falls below a predefined ratio, a slope in the opposite direction is generated, representing a smaller platform. Conversely, when the platform size exceeds the predefined ratio, a larger platform is provided.

Discrete obstacle terrain: In the design of terrains with discrete obstacles, the number of rectangles increases with difficulty. The minimum and maximum sizes of each obstacle are 1.0 and 2.0, respectively, while the platform size is maintained at 3.0 to ensure the robot can navigate obstacles of varying dimensions.

Smooth slope terrain: In the design of smooth slope terrain, the minimum and maximum elevation values define the vertical range, with a minimum of 0.0 and a maximum of 0.1. The step parameter regulates the interval of elevation changes, which is set at 0.15 to ensure discrete variations in the terrain's undulation. The downsampling factor is set to 0.2, controlling the terrain's smoothness and resolution; smaller values yield smoother terrain.

A. Simulation

In this study, we validated the proposed RL control framework using the Webots simulation platform to quantify the robot's performance in terrain adaptability and height adjustment. The validation protocol followed the training curriculum, covering four terrain types: stairs, smooth slopes, rough slopes, and discrete obstacles, with 10 difficulty levels set for each terrain type. For each terrain and difficulty level, randomized trials were conducted by varying initial poses, terrain noise, and random seed parameters to ensure a thorough evaluation. This validated the importance of terrain information, as shown in Fig.2, which highlighted that privileged information was critical for successful learning. Fig.6 illustrates the high-fidelity simulation of the Tron1 robot in Webots.

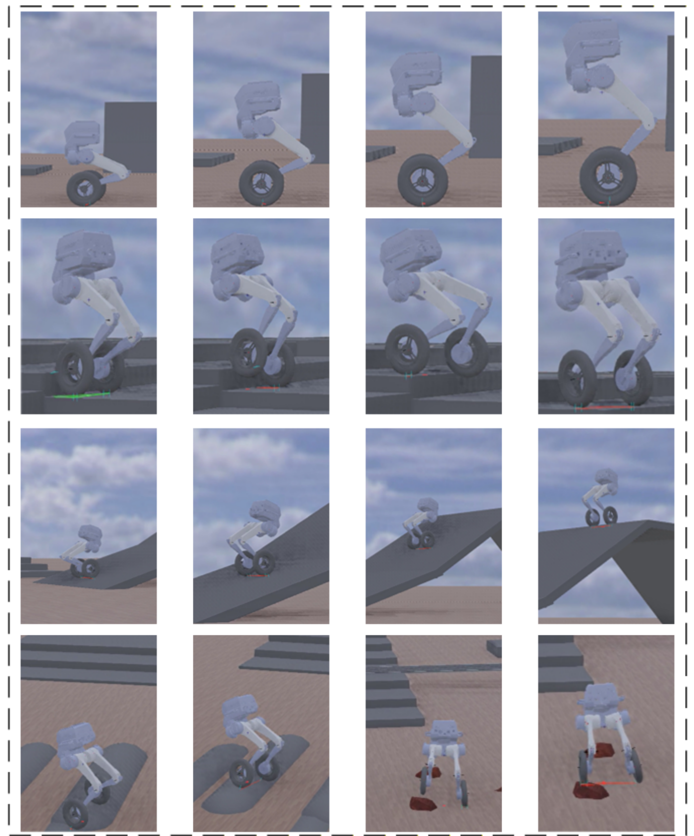

Fig. 6. Robots undergo high-assurance simulation verification in Webots for the four designed terrains. Details can be found in the supplementary video.

As shown in the second row of Fig.6, the robot navigated steps with a height of $10cm$, which confirmed that the robot can traverse such steps using a velocity-triggered strategy in most cases. The third row of robots in Fig.6 demonstrated stability and a high traversal success rate on smooth slopes. It effectively adjusted height and maintained stability under moderate slopes and variable speed commands. The last row of Fig.6 illustrated that the robot faced challenges in complex terrain featuring irregular slopes and interspersed obstacles. Nevertheless, it achieved high success rates traversing such terrain, actively adjusting its posture through height control interpolation to enhance adaptability across diverse surfaces.

Active linear interpolation and velocity-triggered leg-lifting were two critical techniques in this framework. By combining height-joint angle mapping with active interpolation, the robot was able to adjust its height across diverse terrains, avoiding inappropriate posture adjustments. Simultaneously, the velocity-triggered leg-lifting strategy proactively initiated leg-lifting actions based on discrepancies between wheel speed, body velocity, and forward movement commands, ensuring

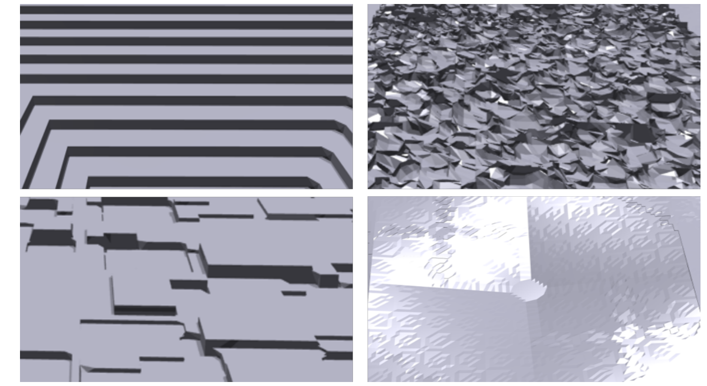

图5. 第一行从左至右：楼梯地形、离散障碍地形。第二行：粗糙斜坡、平滑地形。

叠加到地形上。生成的噪声增强了地形的变化，提高了其真实感。

楼梯地形：在楼梯地形设计中，台阶高度随难度增加而增加。根据机器人训练表现，台阶高度在5厘米至10厘米之间调整。台阶宽度固定为0.35，每个独立平台尺寸设为2.0米。这模拟了楼梯倾斜度，使得楼梯地形能够在单一函数内呈现。

粗糙斜坡地形：在粗糙斜坡地形设计中，坡度随难度增加而调整。难度由独立平台的尺寸控制。当平台尺寸低于预设比例时，会生成一个相反方向的坡度，代表较小的平台。反之，当平台尺寸超过预设比例时，则提供较大的平台。

离散障碍地形：在离散障碍地形设计中，矩形数量随难度增加而增加。每个障碍物的最小尺寸和最大尺寸分别为1.0和2.0，而平台尺寸保持为3.0，以确保机器人能够穿越不同尺寸的障碍物。

平滑斜坡地形：在平滑斜坡地形的设计中，最小和最大高程值定义了垂直范围，最小值为0.0，最大值为0.1。步长参数调节高程变化的间隔，设置为0.15以确保地形起伏的离散变化。降采样因子设置为0.2，控制地形的平滑度和分辨率；较小的值会产生更平滑的地形。

A. 仿真

在本研究中，我们利用Webots仿真平台验证了所提出的强化学习控制框架，以量化机器人在地形适应性和高度调节方面的性能。验证方案遵循训练课程，涵盖了四种地形类型：楼梯、平滑斜坡、粗糙斜坡和离散障碍物，每种地形类型设置了10个难度等级。针对每种地形和难度等级，通过改变初始位姿、地形噪声和随机种子参数进行随机试验，以确保评估的全面性。这验证了地形信息的重要性，如图2所示，该图强调了特权信息对于成功学习至关重要。图6 展示了Tron1机器人在Webots中的高保真仿真。

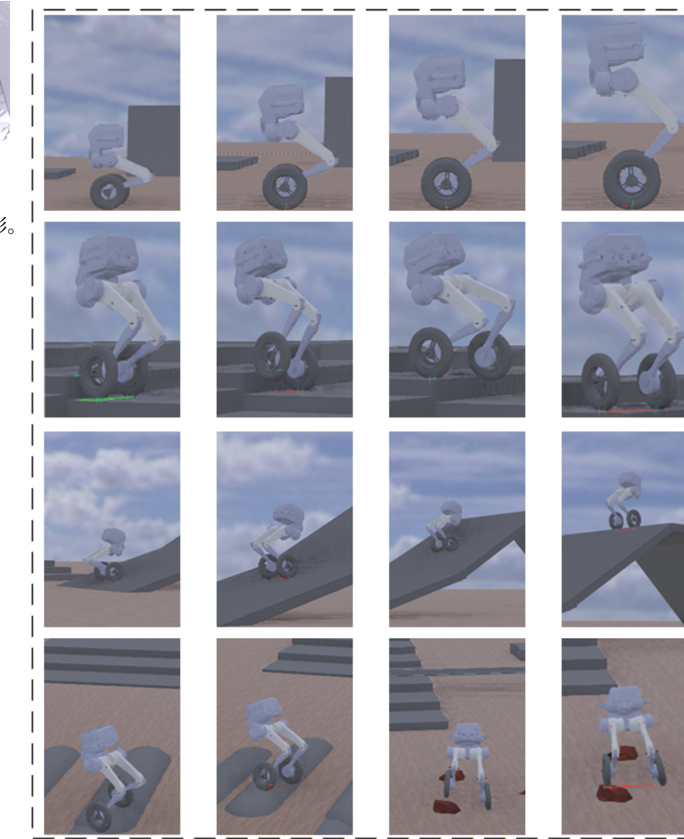

图6. 机器人在Webots中对四种设计地形进行高置信度仿真验证。详情请参见补充视频。

如图6第二行所示，机器人成功通过了高度为10厘米的台阶，这证实了机器人在大多数情况下能够使用速度触发策略穿越此类台阶。图6第三行的机器人 展示了在平滑斜坡上的稳定性和高穿越成功率。它能够在中等坡度和可变速度指令下有效调整高度并保持稳定性。图6最后一行表明，机器人在包含不规则斜坡和散布障碍物的复杂地形中面临挑战。尽管如此，它仍能以高成功率穿越此类地形，并通过高度控制插值主动调整姿态，以增强对不同表面的适应性。

主动线性插值和速度触发的抬腿是该框架中的两项关键技术。通过将高度-关节角度映射与主动插值相结合，机器人能够在多样化地形上调整其高度，避免不适当的姿态调整。同时，速度触发的抬腿策略基于轮速、机体速度与前进运动指令之间的差异，主动启动抬升动作，确保

the robot could navigate obstacles smoothly.

The simulations demonstrated that the RL-based control framework effectively enabled the robot to adapt to diverse terrains while maintaining precise height regulation. The study further indicated that the robot was able to transfer behaviors learned in simulation to real-world environments, a capability subsequently validated through experiments on the Tron1.

B. Real-world Experiments

We deployed the proposed RL control framework on the 8-Degree of Freedom(8-DoF) wheeled robot Tron1. To validate the framework's performance in real-world environments, we designed diverse scenarios to test its traversal capability and posture adjustment capability.

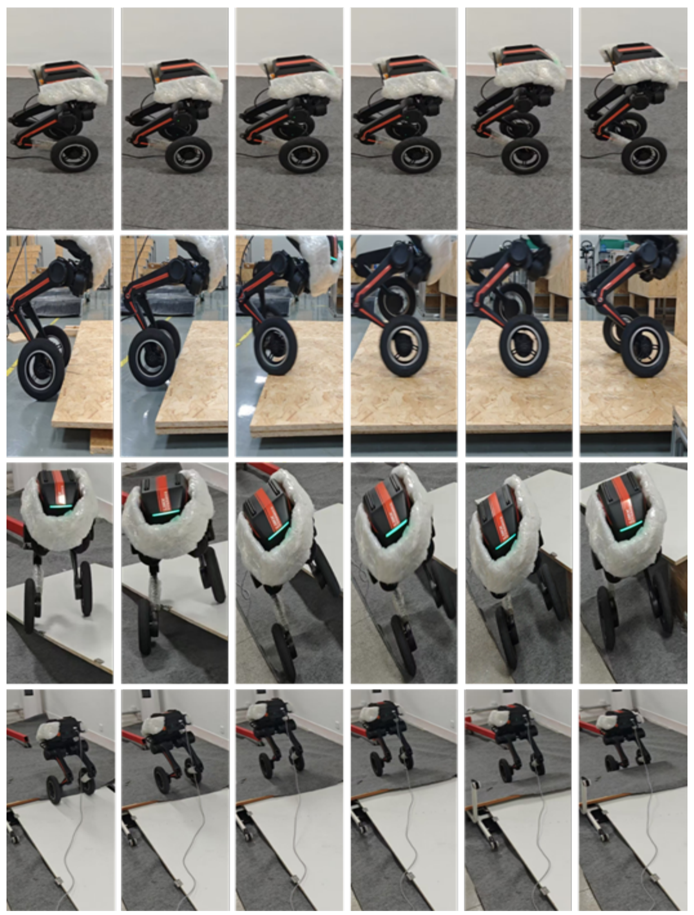

Fig. 7. Snapshots of the real world experiments. https://drive.google.com/file/d/1FsoPUZlac7x1-Ce8O_E8-E4OALCyg5IT/view?usp=sharing

The first row of Fig.7 clearly demonstrated that, while ensuring the robot's mobility, its height could be adjusted through preset height values combined with active linear interpolation. This validated the effectiveness of active interpolation height control. The second row of Fig.7 demonstrated that when encountering a 10*cm* step, the robot could infer the presence of an obstacle based on the tri-source velocity difference. Using a velocity-triggered strategy and weights calculated according to obstruction intensity, the system executed a leg-lifting strategy by rewarding upward velocity when the foot's vertical velocity is positive, ensuring traversal capability.

After verifying its posture adjustment and terrain traversal ability, we suddenly issued a turning command, causing it to fall backward during the slope-climbing period of the robot. We observed that when the robot's right leg landed while the left leg remained suspended, it could still maintain balance by continuing to receive forward movement commands, as shown in the third row of Fig.7. The last row of Fig.7 shows that when traversing dynamic slopes, relative slippage occurred between the wheels and the platform. Upon reaching the slope's apex, issuing a reverse command caused the platform to dynamically oscillate, yet the robot maintained both its traversal capability and equilibrium. This demonstrates that the network exhibits exceptionally strong generalization capabilities.

The experimental results validated the effectiveness and practicality of our approach in real-world settings, confirming the success of this strategy.

IV. Conclusion

In summary, we trained a velocity-triggered control strategy for the B-WLR. By integrating active interpolation height control and tri-source triggered leg-lifting, this strategy enables the robot to navigate complex terrains such as stairs and slopes without relying on external sensors. Despite the encouraging results achieved in this study, several limitations remain to be addressed. Currently, the robot adjusts its height using six interpolation points, resulting in slightly lower precision. Covering all data points with the current method would require excessive preliminary effort.

Future work will replace the existing interpolation method with precise inverse kinematic calculations to provide more accurate joint angle references, thereby enhancing posture adjustment capabilities.

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机器人能够平稳地穿越障碍物。

仿真结果表明，基于强化学习的控制框架有效使机器人能够适应多样化地形，同时保持精确的高度调节。研究进一步指出，机器人能够将在仿真中习得的行为迁移至真实环境，这一能力随后通过Tron1上的实验得到了验证。

B. 真实世界实验

我们将所提出的强化学习控制框架部署在8自由度轮式机器人Tron1上。为验证该框架在真实环境中的性能，我们设计了多种场景来测试其越障能力和姿态调整能力。

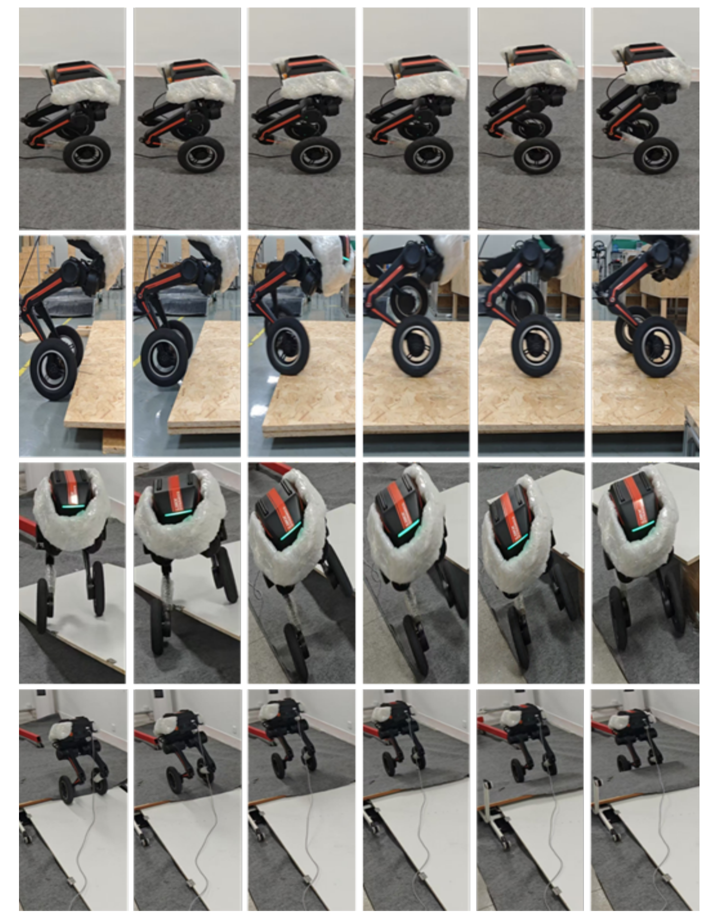

图7. 真实世界实验快照。https://drive.google.com/file/ d/1FsoPUZlac7x1-Ce8O E8-E4OALCyg5IT/view?usp=共享

图7的第一行清晰地表明，在确保机器人机动性的同时，其高度可以通过预设高度值与主动线性插值相结合的方式进行调节。这验证了主动插值高度控制的有效性。图7的第二行展示了当遇到一个10厘米高的台阶时，机器人能够基于三源速度差推断出障碍物的存在。通过使用速度触发策略以及根据障碍强度计算出的权重，系统执行了抬腿策略，即在足部垂直速度为正时奖励向上的速度，从而确保越障能力。

在验证其姿态调整与地形越障能力后，我们突然发出转向指令，导致机器人在爬坡过程中向后倾倒。我们观察到，当机器人右腿着地而左腿悬空时，通过持续接收前进指令，它仍能保持平衡，如图7第三行所示。7。图7最后一行显示，在穿越动态斜坡时，轮子与平台之间发生了相对滑移。到达坡顶后，发出反向指令导致平台动态振荡，但机器人仍保持了越障能力和平衡状态。这表明该网络展现出极强的泛化能力。

实验结果验证了该方法在实际环境中的有效性和实用性，证实了该策略的成功。

IV. 结论

总之，我们为B-WLR训练了一种速度触发控制策略。通过集成主动插值高度控制和三源触发抬腿机制，该策略使机器人能够在无需依赖外部传感器的情况下，在楼梯和斜坡等复杂地形中导航。尽管本研究取得了令人鼓舞的成果，但仍存在若干局限性有待解决。目前，机器人使用六个插值点调整其高度，导致精度略低。用当前方法覆盖所有数据点需要过多的前期工作。

未来的工作将用精确的逆运动学计算取代现有的插值方法，以提供更准确的关节角度参考，从而增强姿态调整能力。

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